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Experiment Name: K-Means Clustering

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

df=pd.read_csv('Mall_Customers.csv')

df.info()
```

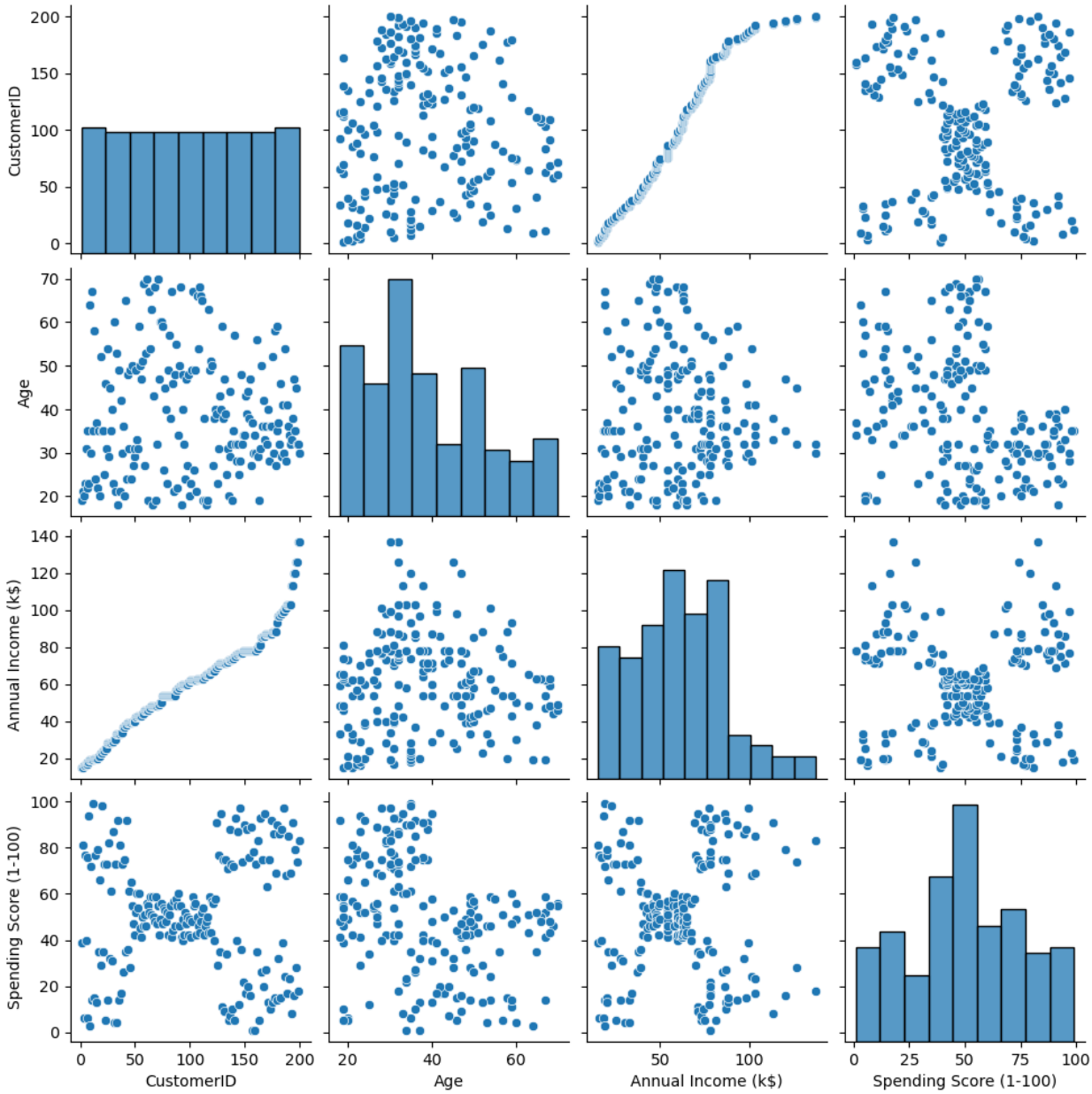
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype  
---  ------                  --------------  -----  
 0   CustomerID              200 non-null    int64  
 1   Gender                  200 non-null    object 
 2   Age                     200 non-null    int64  
 3   Annual Income (k$)      200 non-null    int64  
 4   Spending Score (1-100)  200 non-null    int64  
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

```
df.head()
```

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
sns.pairplot(df)
```

```
<seaborn.axisgrid.PairGrid at 0x1f67ce53230>
```




```
features=df.iloc[:,[3,4]].values

from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
KMeans(n_clusters=5)

C:\Users\swath\anaconda3\Lib\site-packages\sklearn\cluster\
_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on
Windows with MKL, when there are less chunks than available threads.
You can avoid it by setting the environment variable
```

```
OMP_NUM_THREADS=1.
  warnings.warn(

KMeans(n_clusters=5)

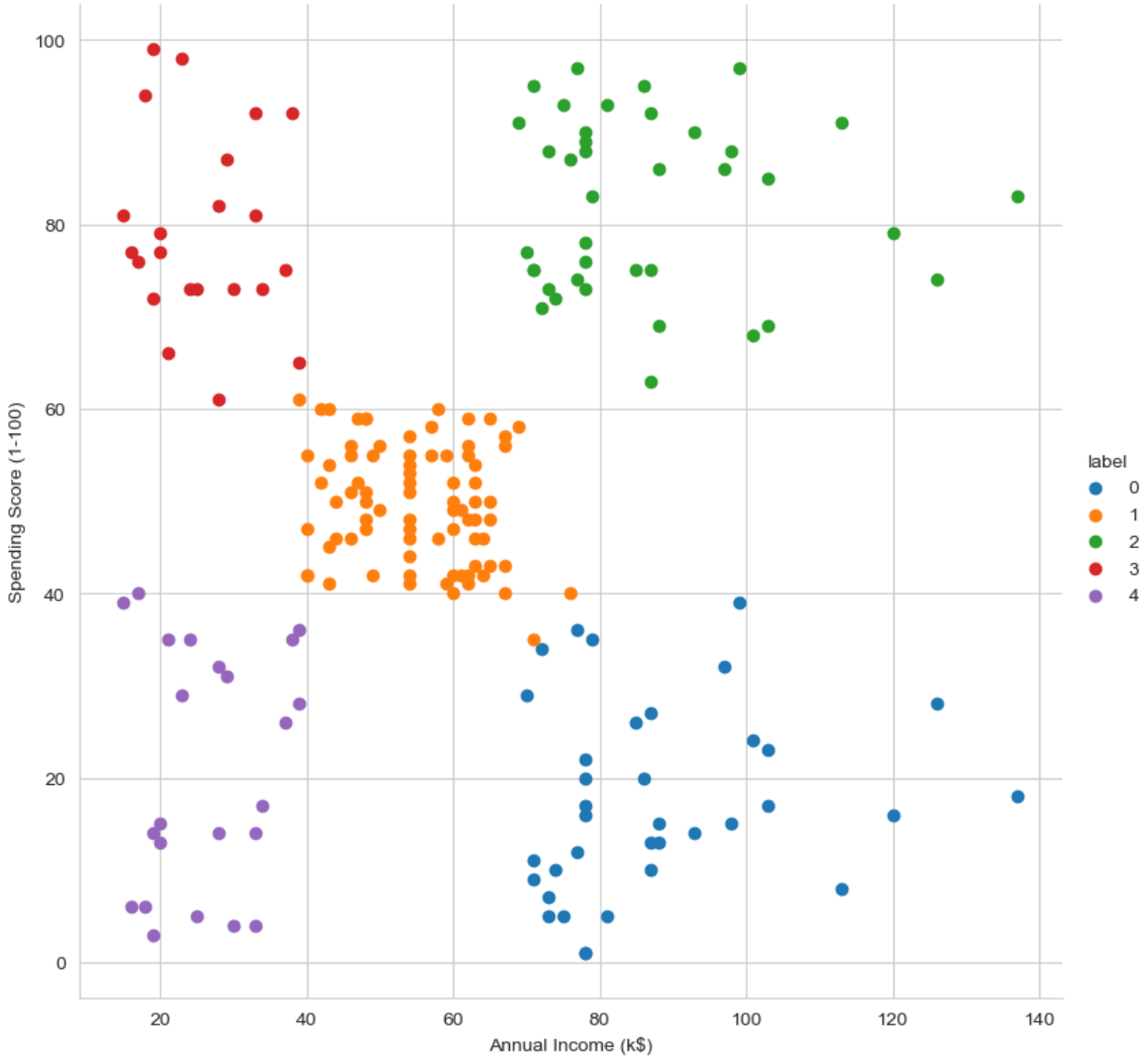
Final=df.iloc[:,[3,4]]
Final['label']=model.predict(features)
Final.head()

C:\Users\swath\AppData\Local\Temp\ipykernel_5828\470183701.py:2:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation:
https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
  Final['label']=model.predict(features)

   Annual Income (k$)  Spending Score (1-100)  label
0                  15                      39      4
1                  15                      81      3
2                  16                       6      4
3                  16                      77      3
4                  17                      40      4

sns.set_style("whitegrid")
sns.FacetGrid(Final,hue="label",height=8) \
.map(plt.scatter,"Annual Income (k$)", "Spending Score (1-100)") \
.add_legend();
plt.show()
```




```
features_el=df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss=[]
for i in range(1,10):
    model=KMeans(n_clusters=i)
model.fit(features_el)
wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```

```
  Cell In[11], line 5
    model=KMeans(n_clusters=i)
    ^
IndentationError: expected an indented block after 'for' statement on
line 4
```